

A strict gap between square isolation and domination in the six-dimensional cube

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Abstract

For a graph G , let $\iota(G, Q_2)$ be the least size of a vertex set whose closed neighborhood meets every subgraph isomorphic to the square Q_2 . We give an exact computer-assisted proof that $\iota(Q_6, Q_2) = 5$, whereas $\gamma(Q_4) = 4$. This disproves the equality $\iota(Q_n, Q_2) = \gamma(Q_{n-2})$ for all integers $n \geq 3$. The lower bound checks all 635 376 four-vertex subsets of Q_6 ; each leaves at least six coordinate squares. An explicit five-vertex set gives the matching upper bound. The projection criterion, witnesses, and complete executable enumeration are provided.

1 Statement and context

The hypercube Q_d has vertex set $\{0, 1\}^d$, with two vertices adjacent when their Hamming distance is one. For $S \subseteq V(G)$, write $N_G[S] = \bigcup_{s \in S} (\{s\} \cup N_G(s))$. A set S is Q_2 -isolating if the induced graph $G[V(G) \setminus N_G[S]]$ contains no subgraph isomorphic to Q_2 . Copies are arbitrary subgraphs; the definition does not impose an additional induced-copy restriction. The minimum size of such a set is $\iota(G, Q_2)$. The domination number $\gamma(G)$ is the least size of D with $N_G[D] = V(G)$.

The problem motivating this note is

$$\iota(Q_n, Q_2) = \gamma(Q_{n-2}) \quad \text{for every integer } n \geq 3. \quad (1)$$

The supplied source record attributes this question to the $k = 2$ case of Brešar and Rall [1, Problem 2], alongside their reported inequality $\iota(Q_n, Q_k) \geq \gamma(Q_{n-k})$ for $0 < k < n$ [1, Eq. (1)]. The contribution here is a strict gap at $(n, k) = (6, 2)$ and the exact isolation value at that point. The argument below proves everything it uses directly.

Theorem 1. *For the hypercube graphs with the definitions above,*

$$\iota(Q_6, Q_2) = 5 > 4 = \gamma(Q_4).$$

In particular, (1) is false.

Related exact finite certification in discrete cube models appears in Zeng, Liu, Ratnavelu, and Huat [2]; this is adjacent computational methodology, with different extremal parameters. Neither reference is a premise of Theorem 1. We make no claim of priority, of a first failing dimension, or of an isolation value in dimensions above six. The primary text and current version of the motivating reference were not independently retrieved for this note; its attribution follows the supplied source record.

2 Squares and projections

Number the coordinates of Q_6 from 0 to 5. Identify a vertex $(v_0, \dots, v_5)$ with $\sum_{i=0}^5 v_i 2^i$. For $I \subseteq \{0, \dots, 5\}$ of size two, let $J = I^c$. The projection p_J retains the coordinates of J in increasing order, with the first retained coordinate again the least significant bit. Write $F_{I,x}$ for the face whose free coordinates are I and whose fixed coordinates have projected label $x \in \{0, \dots, 15\}$.

Lemma 1. *Every subgraph of Q_6 isomorphic to Q_2 is a coordinate two-face. Moreover, for $S \subseteq V(Q_6)$,*

$$F_{I,x} \cap N_{Q_6}[S] = \emptyset \iff x \notin N_{Q_4}[p_J(S)]. \quad (2)$$

Thus S is Q_2 -isolating if and only if all fifteen sets $p_J(S)$ dominate Q_4 .

Proof. Follow the four edges of a simple four-cycle and record their flipped coordinates. Each coordinate occurs an even number of times, because the walk closes. Consecutive flips cannot be equal, including across the end of the cyclic list, since that would repeat a vertex. Consequently exactly two coordinates alternate. The cycle is the coordinate face on those two coordinates. Conversely, every such face is a four-cycle, and its diagonals are not cube edges.

For a selected vertex s , the free coordinates of a face vertex can be chosen to agree with s , whereas discrepancies on J are forced. Hence

$$\min_{v \in F_{I,x}} d_H(s, v) = d_H(p_J(s), x).$$

The face meets $N_{Q_6}[S]$ exactly when this minimum is at most one for some $s \in S$, proving (2). The first part of the lemma then gives the isolation criterion. $\square$

Let $r(S)$ count the coordinate two-faces remaining after deletion of $N_{Q_6}[S]$. Each face has a unique free pair and fixed label, so Lemma 1 yields

$$r(S) = \sum_{\substack{I \subseteq \{0, \dots, 5\} \\ |I|=2}} (16 - |N_{Q_4}[p_{I^c}(S)]|). \quad (3)$$

There are $\binom{6}{2} 16 = 240$ faces in total. Coincident projected labels cause no difficulty because neighborhoods are combined by union.

3 The exact finite certificates

3.1 Excluding four selected vertices

The full enumeration of (3) gives

$$\min_{\substack{S \subseteq \{0, \dots, 63\} \\ |S|=4}} r(S) = 6. \quad (4)$$

One attaining set is $\{0, 3, 61, 62\}$. Appendix A contains the complete program used to reproduce this finite calculation. Its loop visits every increasing quadruple $0 \leq a < b < c < d \leq 63$, exactly $\binom{64}{4} = 635\,376$ sets. It uses no symmetry reduction or numerical optimization.

For clarity, a subset of Q_4 is encoded by a 16-bit integer, with bit x indicating membership of label x . The closed neighborhood of y is encoded by

$$B(y) = 2^y \text{ OR } \bigvee_{j=0}^3 2^{y \text{ XOR } 2^j}.$$

For each free pair, the program precomputes $B(p_{I^c}(v))$ for all 64 vertices. The union of four such masks encodes precisely the dominated labels in that projection. Subtracting its population count from 16 gives the corresponding summand in (3). All operations are exact integer operations. This explains both the checked predicate and the completeness of the finite domain underlying (4).

An additional checker reconstructed the cube adjacency and all 240 four-cycles, then counted the surviving cycles directly for every one of the 635 376 four-sets. Those counts agreed with the projection calculation. A full certificate also records one surviving face for each four-set; all sixteen distances between its four vertices and the four selected vertices were checked to be at least two.

If $S \subseteq T$, then $N[S] \subseteq N[T]$, so every superset of an isolating set is isolating. Every set of size less than four can be extended to a four-set in Q_6 . Since (4) excludes all four-sets, it also excludes all smaller sets, including the empty set. Therefore $\iota(Q_6, Q_2) \geq 5$.

3.2 A five-vertex isolating set

Consider

$$S_* = \{0, 3, 5, 57, 62\} = \{000000, 000011, 000101, 111001, 111110\}_2,$$

where displayed binary strings have their most significant bit first. Its projections fall into the four groups in Table 1. For each listed projected set, the union of its closed neighborhoods in Q_4 is $\{0, \dots, 15\}$: take each label itself and the four labels obtained by toggling a single bit. The appendix program checks each of the fifteen projections separately. By Lemma 1, S_* is isolating, proving $\iota(Q_6, Q_2) \leq 5$.

Free coordinate pairs I	Distinct labels of $p_{I^c}(S_*)$
$(0, 1), (0, 2), (1, 2)$	$\{0, 1, 14, 15\}$
$(0, 3), (0, 4), (0, 5)$	$\{0, 1, 2, 12, 15\}$
$(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)$	$\{0, 1, 3, 13, 14\}$
$(3, 4), (3, 5), (4, 5)$	$\{0, 3, 5, 9, 14\}$

Table 1: Every projected set dominates Q_4 . Coordinate and label conventions are those of Lemma 1.

3.3 Domination of the four-dimensional cube

Each closed neighborhood in Q_4 has five vertices. Three selected vertices therefore cover at most fifteen of the sixteen vertices, so $\gamma(Q_4) \geq 4$. For the upper bound take $D = \{0, 1, 14, 15\}$. Vertices of Hamming weight at most one belong to $N[0]$, and those of weight at least three belong to $N[15]$. A weight-two vertex with bit 0 equal to one is adjacent to 1; one with bit 0 equal to zero is adjacent to 14. Thus D dominates and $\gamma(Q_4) = 4$. Together with the two isolation bounds, this proves Theorem 1.

Computational scope and assistance. The lower bound is an exact finite computer-assisted proof with the ordinary trusted-runtime assumptions of Python integer computation; it is not a proof-assistant formalization. AI assistants contributed to proof development, programming, verification, and manuscript preparation. The separate verification pass reconstructed the graph checks in a fresh model context; this does not constitute external human peer review. The printed program reproduces the central finite claims without input data, random seeds, or third-party libraries.

A Executable enumeration

Save the listing as `verify.py` and run `python3 verify.py`. The complete output is

```
four_sets = 635376
minimum_surviving_faces = 6
five_set_isolates = True; gamma_Q4 = 4
```

The count and minimum concern the entire four-set domain; the final line checks the explicit five-set and the domination certificate.

```
from itertools import combinations

# Integer labels use bit i for coordinate i, with bit 0 least significant.
balls = [(1 << x) | sum(1 << (x ^ (1 << j)) for j in range(4))
          for x in range(16)]
pairs = list(combinations(range(6), 2))
kept = [[j for j in range(6) if j not in I] for I in pairs]
rows = [[balls[sum((v >> j) & 1) << k
                for k, j in enumerate(J)]] for v in range(64)]
        for J in kept]
pop = [bin(x).count("1") for x in range(65536)]

minimum, count = 240, 0
for a, b, c, d in combinations(range(64), 4):
    r = sum(16 - pop[row[a] | row[b] | row[c] | row[d]]
            for row in rows)
    minimum = min(minimum, r)
    count += 1
if (count, minimum) != (635376, 6):
    raise RuntimeError((count, minimum))

S = (0, 3, 5, 57, 62)
for row in rows:
    covered = 0
    for v in S:
        covered |= row[v]
    if covered != 65535:
        raise RuntimeError("Five-set fails a projection")

covered = 0
for v in (0, 1, 14, 15):
    covered |= balls[v]
if covered != 65535 or 3 * 5 >= 16:
    raise RuntimeError("Domination certificate fails")
print("four_sets =", count)
print("minimum_surviving_faces =", minimum)
print("five_set_isolates = True; gamma_Q4 = 4")
```

References

- [1] B. Brešar and D. F. Rall, *On the isolation numbers in graph products*, arXiv:2608.25752v1 (2026). <https://arxiv.org/abs/2608.25752v1>.
- [2] Z. Zeng, H. Liu, K. Ratnavelu, and O. S. Huat, *Exact Finite Certificates for the Four-Dimensional Ternary Borsuk Problem and Five-Dimensional Ternary Kissing Codes*, Preprints (2026). <https://doi.org/10.20944/preprints202608.1658.v1>.